# CONCEPT NOTE

**Project Title**:
Agora-Link: A Privacy-Preserving AI Compliance Gateway

## 1. Background/Context

The rapid adoption of Web3 technology has created a significant friction point: the "Compliance Gap." Traditional financial systems require strict adherence to Know Your Customer (KYC) and Anti-Money Laundering (AML) regulations. However, public blockchains are inherently transparent, meaning that traditional KYC methods often force users to sacrifice their privacy to meet legal standards. This conflict prevents large-scale institutional adoption of crypto-to-card payment systems.

Addressing this is crucial for AgoraX's mission to bridge traditional payments with decentralized infrastructure. By creating a system that satisfies regulators without exposing sensitive user data, we can build a truly sustainable and secure financial bridge.

- **Theme 3 (Web3 Adoption)**: Creating a bridge for card-to-crypto payments.
- **Theme 5 (AI for Compliance)**: Using ML for automated AML/KYC.
- **Theme 1 (Security)**: Ensuring the bridge is cryptographically secure.

## 2. Objectives

The primary objective of this project is to develop a research prototype for a payment gateway that integrates:

- **Automated AML Risk Scoring:** Using AI/ML to detect fraudulent behavioral patterns in real-time.

- **Privacy-Preserving KYC:** Utilizing concepts from Garbled Circuits (like COTI V2) to verify identity without on-chain data leakage.

- **Secure Web2-to-Web3 Settlement:** Ensuring crypto tokens are only released once both card payment and compliance checks are validated.

## 3. Target Audience/Beneficiaries

- **Primary Beneficiaries:** Web3 developers and startups (like AgoraX) looking for compliant payment rails; everyday users who want to use credit cards for crypto without losing privacy.

- **Secondary Beneficiaries:** Financial regulators and traditional banking institutions who require transparency and safety in blockchain transactions.

## 4. Key Activities

- AI Engine Development: Building a Python-based engine to analyze transaction velocity and wallet blacklists.
- Privacy Layer Simulation: Researching Garbled Circuit logic for "private" identity verification.
- Full-Stack Integration: Developing a glassmorphic React UI and a Node.js backend to coordinate between AI and the blockchain.
- EIP-712 Implementation: Utilizing typed data signing to ensure compliance signatures are secure and unique

## 5. Technical Stack (Innovation & Proactivity)

| Layer | Technology | Reason for Selection |
|---|---|---|
| **Frontend** | React.js + Tailwind CSS | For a responsive, "Glassmorphic" premium user experience. |
| **Backend** | Node.js (Express) | Efficient handling of payment webhooks and asynchronous blockchain calls. |
| **AI Engine** | Python (Scikit-learn) | Industry standard for anomaly detection and behavior-based AML logic. |
| **Blockchain** | Solidity + Hardhat | To build and test secure, production-ready settlement smart contracts. |

## 6. Expected Outcomes

The project is expected to deliver a working Proof of Concept (POC) that demonstrates:

- **Data Minimization:** A 95% reduction in public exposure of sensitive user identity data.
- **Proactive Compliance:** A functional dashboard that flags high-risk behavior before transactions occur.
- **Seamless Bridge:** A working POC demonstrating an end-to-end card-to-wallet settlement flow.

## 7. Timeline

- **Phase 1 (Days 1–4):** Initial research into Garbled Circuits and development of the AI AML logic.
- **Phase 2 (Days 5–10):** Implementation of the Node.js bridge and React frontend; Smart Contract logic development.
- **Phase 3 (Days 11–14):** Final integration, testing for edge cases, and completion of technical documentation.

## 8. Time Estimate

- **Research & AI Model Training:** 30 Man-Hours
- **Full-Stack & Blockchain Development:** 40 Man-Hours
- **Testing & Documentation:** 10 Man-Hours
- **Total Resource Allocation:** 80 Man-Hours

## 9. Conclusion

This project offers a tangible, technical solution to the most pressing challenge in Web3: making decentralized finance (DeFi) safe and compliant without destroying the core value of privacy. By integrating AI-driven AML and ZK-ready privacy layers, "Agora-Link" provides a roadmap for the future of regulated global payments. Supporting this initiative demonstrates a commitment to building a more secure and inclusive digital economy.

## 10. Contact Information


- **Lead Developer:** Harshdeep Sharma
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- **Specialization:** B.Tech CSE (AI & ML)